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10 / 10 submitted

Q1. Selection Sort

Score: 10.00 TC: 3/3 passed

CODE

```
def selection_sort(arr):
    for i in range(len(arr)):
        min_idx = i
        for j in range(i+1, len(arr)):
            if arr[j] < arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr

arr = list(map(int, input().split()))
print(selection_sort(arr))
```

INPUT 23 78 45 8 32 56

OUTPUT [8, 23, 32, 45, 56, 78]

Q2. Insertion Sort

Score: 10.00 TC: 3/3 passed

CODE

```
arr = list(map(int, input().split()))

for i in range(1, len(arr)):
    key = arr[i]
    j = i - 1

    while j >= 0 and arr[j] > key:
        arr[j + 1] = arr[j]
        j -= 1

    arr[j + 1] = key

print(arr)
```

INPUT 8 5 7 1 9 3

OUTPUT [1, 3, 5, 7, 8, 9]

Q3. Merge Sort

Score: 10.00 TC: 3/3 passed

CODE

```
def merge_sort(arr):
    if len(arr) > 1:
        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)

        i = j = k = 0
        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1

        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1

        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1

    return arr

arr = list(map(int, input().split()))
print(merge_sort(arr))
```

INPUT 38 27 43 3 9 82 10

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Q4. Diagonal Matrix

Score: 10.00 TC: 3/3 passed

CODE

```
s = input().strip()

s = s.replace("[", "").replace("]", "")
numbers = list(map(int, s.split(",")))

import math

n = math.isqrt(len(numbers))

matrix = []

for i in range(n):
    matrix.append(numbers[i*n : (i+1)*n])

is_diagonal = True

for i in range(n):
    for j in range(n):
        if i != j and matrix[i][j] != 0:
            is_diagonal = False

if is_diagonal:
    print("Diagonal Matrix")
else:
    print("Not Diagonal Matrix")
```

INPUT OUTPUT

Q5. Row and Column Sums

Score: 6.67 TC: 2/3 passed

CODE

```
s = input().replace(" ", "")

rows = s[2:-2].split(",")

matrix = [list(map(int, r.split(","))) for r in rows]

row_sum = [sum(row) for row in matrix]

col_sum = [sum(matrix[r][c] for r in range(len(matrix))) for c in range(len(matrix[0]))]

print("Row=", row_sum)

print("Col=", col_sum)
```

INPUT OUTPUT

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Q6. Maximum Subarray Sum

Score: 6.67 TC: 2/3 passed

CODE

```
s = input().strip()
s = s.replace("[", "").replace("]", "").replace(" ", " ")
arr = [int(x) for x in s.split()]
max_sum = arr[0]
current_sum = arr[0]
for x in arr[1:]:
    current_sum = max(x, current_sum + x)
    max_sum = max(max_sum, current_sum)
print(max_sum)
```

INPUT OUTPUT

111 / 122



Q7. Common Elements Among Three Lists

Score: 10.00 TC: 3/3 passed

CODE

```
def common_elements(list1, list2, list3):
    return list(set(list1) & set(list2) & set(list3))

lists = input().split(';')
list1 = list(map(int, lists[0].split()))
list2 = list(map(int, lists[1].split()))
list3 = list(map(int, lists[2].split()))

print(common_elements(list1, list2, list3))
```

INPUT OUTPUT

Q8. Matrix Multiplication

Score: 6.67 TC: 2/3 passed

CODE

```

s = input().strip()

if " B=" in s or (" " in s and "=" in s):
    parts = s.split(" B=") if " B=" in s else s.split()
    s1 = parts[0]
    s2 = parts[1]
else:
    s1 = s
    try:
        s2 = input().strip()
    except EOFError:
        s2 = ""

if "=" in s1:
    s1 = s1.split("=", 1)[1].strip()

if "=" in s2:
    s2 = s2.split("=", 1)[1].strip()

def parse_matrix(s):
    s = s.replace(" ", "")
    s = s[1:-1]
    rows = []
    curr = ""
    inside = False

    for char in s:
        if char == '[':
            inside = True
            curr = ""
        elif char == ']':
            inside = False
            if curr:
                rows.append([int(x) for x in curr.split(",")])
            elif inside:
                curr += char

    return rows

A = parse_matrix(s1)
B = parse_matrix(s2)

rows_A = len(A)
cols_A = len(A[0])
cols_B = len(B[0])

result = [[0] * cols_B for _ in range(rows_A)]

for i in range(rows_A):
    for j in range(cols_B):
        for k in range(cols_A):
            result[i][j] += A[i][k] * B[k][j]

rows_str = [str(row) for row in result]

print("[ " + "\n ".join(rows_str) + " ]")

```

INPUT

```
[[1,2],[3,4]]
[[5,6],[7,8]]
```

OUTPUT

```
[[19, 22]
 [43, 50]]
```

Q9. Sort Rotate Min Max

Score: 3.33 TC: 1/3 passed

CODE

```

import sys
import re

input_data = sys.stdin.read().strip()

if input_data:
    if "List=" in input_data or "k=" in input_data:
        list_match = re.search(r'List=s*([,\n]+)', input_data)
        k_match = re.search(r'k=s*(\d+)', input_data)

        if list_match and k_match:
            arr = [int(x) for x in list_match.group(1).strip().split()]
            k = int(k_match.group(1))

            else:
                nums = [int(x) for x in re.findall(r'?\d+', input_data)]
                arr = nums[:-1]
                k = nums[-1]

        else:
            lines = [line.strip() for line in input_data.splitlines() if line.strip()]

            if len(lines) >= 2:
                arr = [int(x) for x in lines[0].split()]
                k = int(lines[1])
            else:
                nums = [int(x) for x in lines[0].split()]
                arr = nums[:-1]
                k = nums[-1]

        sorted_arr = sorted(arr)
        n = len(sorted_arr)

        k = k % n if n > 0 else 0

        rotated_arr = sorted_arr[k:] + sorted_arr[:k]

        mn = sorted_arr[0]
        mx = sorted_arr[-1]

```



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Q10. Sort Matrix Search

Score: 3.33 TC: 1/3 passed

CODE

```

import sys

input_data = sys.stdin.read().strip()

if input_data:
    if "Matrix=" in input_data and "Find=" in input_data:
        matrix_part, find_part = input_data.split("Find=")
        target_val = int(find_part.strip())
        matrix_str = matrix_part.split("Matrix=")[1].strip().rstrip(",")
    else:
        lines = [line.strip() for line in input_data.splitlines() if line.strip()]

        if len(lines) ≥ 2:
            matrix_str = lines[0]
            target_val = int(lines[1])
        else:
            parts = input_data.split()
            matrix_str = parts[0]
            target_val = int(parts[1])

    rows_str = matrix_str.strip("[]").split(",")

    matrix = []

    for row in rows_str:
        clean_row = row.replace("[", "").replace("]", "")
        nums = [int(x.strip()) for x in clean_row.split(",") if x.strip()]
        matrix.append(nums)

    rows = len(matrix)
    cols = len(matrix[0])

    flat_elements = [val for row in matrix for val in row]
    flat_elements.sort()

    sorted_matrix = []
    idx = 0

    for r in range(rows):
        row = []
        for c in range(cols):
            row.append(flat_elements[idx])
            idx += 1
        sorted_matrix.append(row)

    pos = None

    for r in range(rows):
        for c in range(cols):
            if sorted_matrix[r][c] == target_val:
                pos = (r, c)
                break
        if pos:
            break

    if pos:
        formatted_matrix = str(sorted_matrix).replace(" ", "[", "[")
        print(f"Sorted= {formatted_matrix}")
        print(f"Position= {pos}")
    else:
        print("Not Found")

```

INPUT

```

[[5,1],[4,2]]
2

```

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OUTPUT

```

Sorted= [[1, 2],[4, 5]]
Position= (0, 1)

```

